# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

**Creation Date** 12-Apr-2023 **Revision Date** 24-Mar-2024 **Revision Number** 2

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

**Product Description:** **Coulomat AG-Oven reagent for coulometric Karl Fischer titration with oven**
**Cat No. :** **U90019**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Laboratory chemicals.
**Uses advised against** No Information available

### 1.3. Details of the supplier of the safety data sheet

**Company** .

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

**Swiss distributor -** Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
**CHEMTREC** Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

**CLP Classification - Regulation (EC) No 1272/2008**

**Physical hazards**

| | |
|---|---|
| Flammable liquids | Category 3 (H226) |

**Health hazards**

| | |
|---|---|
| Acute oral toxicity | Category 3 (H301) |
| Acute dermal toxicity | Category 4 (H312) |
| Acute Inhalation Toxicity - Vapors | Category 3 (H331) |
| Skin Corrosion/Irritation | Category 1 (H314) B |
| Serious Eye Damage/Eye Irritation | Category 1 (H318) |
| Reproductive Toxicity | Category 1B (H360D) (H361fd) |
| Specific target organ toxicity - (single exposure) | Category 1 (H370) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373) |

**Environmental hazards**
Based on available data, the classification criteria are not met

*Full text of Hazard Statements: see section 16*

## 2.2. Label elements

**Signal Word** **Danger**

**Hazard Statements**

H226 - Flammable liquid and vapor
H312 - Harmful in contact with skin
H314 - Causes severe skin burns and eye damage
H370 - Causes damage to organs
H360D - May damage the unborn child
H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child
H373 - May cause damage to organs through prolonged or repeated exposure
H301 + H331 - Toxic if swallowed or if inhaled

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician

**Additional EU labelling**
Restricted to professional users

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component | CAS No | EC No | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|---|---|---|---|---|
| Propylene carbonate | 108-32-7 | EEC No. 203-572-1 | 38 | Eye Irrit. 2 (H319) |
| Methanol | 67-56-1 | 200-659-6 | 30 | Flam. Liq. 2 (H225)<br>Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>STOT SE 1 (H370) |
| Diethanolamine | 111-42-2 | EEC No. 203-868-0 | 12 | Acute Tox. 4 (H302)<br>Skin Irrit. 2 (H315)<br>Eye Dam. 1 (H318)<br>Repr. 2 (H361fd)<br>STOT RE 2 (H373) |
| Ethylene glycol | 107-21-1 | EEC No. 203-473-3 | 8 | Acute Tox. 4 (H302) |
| Sulfur dioxide | 7446-09-5 | EEC No. 231-195-2 | 7 | Press. Gas (H280)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>STOT SE 1 (H370)<br>Acute Tox. 3 (H331) |
| 1-Imidazole | 288-32-4 | EEC No. 206-019-2 | 3 | Acute Tox. 4 (H302)<br>Skin Corr. 1C (H314)<br>Eye Dam. 1 (H318)<br>Repr. 1B (H360D) |
| Iodized salt | N/A | | 2 | - |

| Component | Specific concentration limits (SCL's) | M-Factor | Component notes |
|---|---|---|---|
| Methanol | STOT Single Exp. 1 :: >= 10<br>STOT Single Exp. 2 :: 3 - < 10 | - | - |

*Full text of Hazard Statements: see section 16*

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

**General Advice** Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact** Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact** Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Ingestion** Do NOT induce vomiting. Call a physician or poison control center immediately.

**Inhalation** If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### 4.3. Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

**Suitable Extinguishing Media**
$CO_2$, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**
No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Hazardous Combustion Products**
Carbon oxides, Nitrogen oxides (NOx), Sulfur oxides, Hydrogen iodide.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Hygiene Measures**
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

**Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany)** Class 3

**Switzerland - Storage of hazardous substances** Storage class - SC 3
https://www.kvu.ch/de/themen/stoffe-und-produkte
https://www.kvu.ch/fr/themes/substances-et-produits
https://www.kvu.ch/it/temi/sostanze-e-prodotti

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

**Exposure limits**
List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union | The United Kingdom | France | Belgium | Spain |
|---|---|---|---|---|---|
| Methanol | TWA: 200 ppm 8 hr<br>TWA: 260 mg/m³ 8 hr<br>Skin | WEL - TWA: 200 ppm TWA; 266 mg/m³ TWA<br>WEL - STEL: 250 ppm STEL; 333 mg/m³ STEL | TWA / VME: 200 ppm (8 heures). restrictive limit<br>TWA / VME: 260 mg/m³ (8 heures). restrictive limit<br>STEL / VLCT: 1000 ppm. restrictive limit<br>STEL / VLCT: 1300 mg/m³. restrictive limit<br>Peau | TWA: 200 ppm 8 uren<br>TWA: 266 mg/m³ 8 uren<br>STEL: 250 ppm 15 minuten<br>STEL: 333 mg/m³ 15 minuten<br>Huid | TWA / VLA-ED: 200 ppm (8 horas)<br>TWA / VLA-ED: 266 mg/m³ (8 horas)<br>Piel |
| Diethanolamine | | | TWA / VME: 3 ppm (8 heures).<br>TWA / VME: 15 mg/m³ (8 heures). | TWA: 0.2 ppm 8 uren<br>TWA: 1 mg/m³ 8 uren<br>Huid | TWA / VLA-ED: 0.2 ppm (8 horas)<br>TWA / VLA-ED: 1 mg/m³ (8 horas)<br>Piel |
| Ethylene glycol | TWA: 20 ppm (8h) | STEL: 40 ppm 15 min | TWA / VME: 20 ppm (8 | TWA: 20 ppm 8 uren | STEL / VLA-EC: 40 ppm |

| | | | | | |
|---|---|---|---|---|---|
| | TWA: 52 mg/m$^3$ (8h)<br>STEL: 40 ppm (15min)<br>STEL: 104 mg/m$^3$ (15min)<br>Skin | STEL: 104 mg/m$^3$ 15 min<br>STEL: 30 mg/m$^3$ 15 min<br>TWA: 10 mg/m$^3$ 8 hr<br>TWA: 20 ppm 8 hr<br>TWA: 52 mg/m$^3$ 8 hr<br>Skin | heures). indicative limit<br>TWA / VME: 52 mg/m$^3$ (8 heures). indicative limit<br>STEL / VLCT: 40 ppm. indicative limit<br>STEL / VLCT: 104 mg/m$^3$. indicative limit<br>Peau | TWA: 52 mg/m$^3$ 8 uren<br>STEL: 40 ppm 15 minuten<br>STEL: 104 mg/m$^3$ 15 minuten<br>Huid | (15 minutos).<br>STEL / VLA-EC: 104 mg/m$^3$ (15 minutos).<br>TWA / VLA-ED: 20 ppm (8 horas)<br>TWA / VLA-ED: 52 mg/m$^3$ (8 horas)<br>Piel |
| Sulfur dioxide | TWA: 1.3 mg/m$^3$ (8h)<br>TWA: 0.5 ppm (8h)<br>STEL: 2.7 mg/m$^3$ (15min)<br>STEL: 1 ppm (15min) | STEL: 1 ppm 15 min<br>STEL: 2.7 mg/m$^3$ 15 min<br>TWA: 0.5 ppm 8 hr<br>TWA: 1.3 mg/m$^3$ 8 hr | TWA / VME: 0.5 ppm (8 heures).<br>TWA / VME: 1.3 mg/m$^3$ (8 heures).<br>STEL / VLCT: 1 ppm. indicative limit<br>STEL / VLCT: 2.7 mg/m$^3$. indicative limit | TWA: 0.5 ppm 8 uren<br>TWA: 1.3 mg/m$^3$ 8 uren<br>STEL: 1 ppm 15 minuten<br>STEL: 2.7 mg/m$^3$ 15 minuten | STEL / VLA-EC: 2 ppm (15 minutos).<br>STEL / VLA-EC: 5.28 mg/m$^3$ (15 minutos).<br>TWA / VLA-ED: 0.5 ppm (8 horas)<br>TWA / VLA-ED: 1.32 mg/m$^3$ (8 horas) |

| Component | Italy | Germany | Portugal | The Netherlands | Finland |
|---|---|---|---|---|---|
| Propylene carbonate | | TWA: 2 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 8.5 mg/m$^3$ (8 Stunden). AGW - exposure factor 1<br>TWA: 2 ppm (8 Stunden). MAK<br>TWA: 8.5 mg/m$^3$ (8 Stunden). MAK<br>Höhepunkt: 2 ppm<br>Höhepunkt: 8.5 mg/m$^3$ | | | |
| Methanol | TWA: 200 ppm 8 ore. Time Weighted Average<br>TWA: 260 mg/m$^3$ 8 ore. Time Weighted Average<br>Pelle | 100 ppm TWA MAK; 130 mg/m$^3$ TWA MAKSkin absorber | STEL: 250 ppm 15 minutos<br>TWA: 200 ppm 8 horas<br>TWA: 260 mg/m$^3$ 8 horas<br>Pele | huid<br>TWA: 133 mg/m$^3$ 8 uren | TWA: 200 ppm 8 tunteina<br>TWA: 270 mg/m$^3$ 8 tunteina<br>STEL: 250 ppm 15 minuutteina<br>STEL: 330 mg/m$^3$ 15 minuutteina<br>Iho |
| Diethanolamine | | TWA: 0.11 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 0.5 mg/m$^3$ (8 Stunden). AGW - exposure factor 1<br>TWA: 1 mg/m$^3$ (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>Höhepunkt: 1 mg/m$^3$<br>Haut | TWA: 1 mg/m$^3$ 8 horas<br>Pele | | TWA: 0.46 ppm 8 tunteina<br>TWA: 2 mg/m$^3$ 8 tunteina<br>Iho |
| Ethylene glycol | TWA: 20 ppm 8 ore. Time Weighted Average<br>TWA: 52 mg/m$^3$ 8 ore. Time Weighted Average<br>STEL: 40 ppm 15 minuti. Short-term<br>STEL: 104 mg/m$^3$ 15 minuti. Short-term<br>Pelle | TWA: 10 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 26 mg/m$^3$ (8 Stunden). AGW - exposure factor 2<br>TWA: 10 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>TWA: 26 mg/m$^3$ (8 Stunden). MAK can occur as vapor and aerosol at the same time | STEL: 40 ppm 15 minutos<br>STEL: 104 mg/m$^3$ 15 minutos<br>Ceiling: 100 mg/m$^3$<br>TWA: 20 ppm 8 horas<br>TWA: 52 mg/m$^3$ 8 horas<br>Pele | huid<br>STEL: 104 mg/m$^3$ 15 minuten<br>TWA: 52 mg/m$^3$ 8 uren<br>TWA: 10 mg/m$^3$ 8 uren | TWA: 20 ppm 8 tunteina<br>TWA: 50 mg/m$^3$ 8 tunteina<br>STEL: 40 ppm 15 minuutteina<br>STEL: 100 mg/m$^3$ 15 minuutteina<br>Iho |

| | | | | | |
|---|---|---|---|---|---|
| | | Höhepunkt: 20 ppm<br>Höhepunkt: 52 mg/m³<br>Haut | | | |
| Sulfur dioxide | TWA: 1.3 mg/m³ 8 ore. Time Weighted Average<br>TWA: 0.5 ppm 8 ore. Time Weighted Average<br>STEL: 2.7 mg/m³ 15 minuti. Short-term<br>STEL: 1 ppm 15 minuti. Short-term | TWA: 1 ppm<br>TWA: 2.5 mg/m³ | STEL: 1 ppm 15 minutos<br>STEL: 2.7 mg/m³ 15 minutos<br>TWA: 0.5 ppm 8 horas<br>TWA: 1.3 mg/m³ 8 horas | STEL: 0.7 mg/m³<br>MAC: 2 ppm<br>MAC: 5 mg/m³ | TWA: 0.5 ppm 8 tunteina<br>TWA: 1.3 mg/m³ 8 tunteina<br>STEL: 1 ppm 15 minuutteina<br>STEL: 2.7 mg/m³ 15 minuutteina |

| Component | Austria | Denmark | Switzerland | Poland | Norway |
|---|---|---|---|---|---|
| Propylene carbonate | | | STEL: 6 ppm 15 Minuten<br>STEL: 25.5 mg/m³ 15 Minuten<br>TWA: 6 ppm 8 Stunden<br>TWA: 25.5 mg/m³ 8 Stunden | | |
| Methanol | Haut<br>MAK-KZGW: 800 ppm 15 Minuten<br>MAK-KZGW: 1040 mg/m³ 15 Minuten<br>MAK-TMW: 200 ppm 8 Stunden<br>MAK-TMW: 260 mg/m³ 8 Stunden | TWA: 200 ppm 8 timer<br>TWA: 260 mg/m³ 8 timer<br>STEL: 400 ppm 15 minutter<br>STEL: 520 mg/m³ 15 minutter<br>Hud | Haut/Peau<br>STEL: 400 ppm 15 Minuten<br>STEL: 520 mg/m³ 15 Minuten<br>TWA: 200 ppm 8 Stunden<br>TWA: 260 mg/m³ 8 Stunden | STEL: 300 mg/m³ 15 minutach<br>TWA: 100 mg/m³ 8 godzinach | TWA: 100 ppm 8 timer<br>TWA: 130 mg/m³ 8 timer<br>STEL: 150 ppm 15 minutter. value calculated<br>STEL: 162.5 mg/m³ 15 minutter. value calculated<br>Hud |
| Diethanolamine | Haut<br>MAK-KZGW: 0.92 ppm 15 Minuten<br>MAK-KZGW: 4 mg/m³ 15 Minuten<br>MAK-TMW: 0.46 ppm 8 Stunden<br>MAK-TMW: 2 mg/m³ 8 Stunden | TWA: 0.46 ppm 8 timer<br>TWA: 2 mg/m³ 8 timer<br>STEL: 0.92 ppm 15 minutter<br>STEL: 4 mg/m³ 15 minutter<br>Hud | Haut/Peau<br>STEL: 1 mg/m³ 15 Minuten<br>TWA: 1 mg/m³ 8 Stunden | TWA: 9 mg/m³ 8 godzinach | TWA: 3 ppm 8 timer<br>TWA: 15 mg/m³ 8 timer<br>STEL: 6 ppm 15 minutter. value calculated<br>STEL: 22.5 mg/m³ 15 minutter. value calculated |
| Ethylene glycol | Haut<br>MAK-KZGW: 20 ppm 15 Minuten<br>MAK-KZGW: 52 mg/m³ 15 Minuten<br>MAK-TMW: 10 ppm 8 Stunden<br>MAK-TMW: 26 mg/m³ 8 Stunden | TWA: 10 ppm 8 timer<br>TWA: 26 mg/m³ 8 timer<br>TWA: 10 mg/m³ 8 timer<br>STEL: 104 mg/m³ 15 minutter<br>STEL: 40 ppm 15 minutter<br>STEL: 20 mg/m³ 15 minutter<br>Hud | Haut/Peau<br>STEL: 20 ppm 15 Minuten<br>STEL: 52 mg/m³ 15 Minuten<br>TWA: 10 ppm 8 Stunden<br>TWA: 26 mg/m³ 8 Stunden | STEL: 50 mg/m³ 15 minutach<br>TWA: 15 mg/m³ 8 godzinach | TWA: 20 ppm 8 timer<br>TWA: 52 mg/m³ 8 timer<br>STEL: 104 mg/m³ 15 minutter. total sum of gas and particulate matter (aerosol) of the substance;value from the regulation<br>STEL: 40 ppm 15 minutter. total sum of gas and particulate matter (aerosol) of the substance;value from the regulation<br>Hud |
| Sulfur dioxide | MAK-KZGW: 1 ppm 15 Minuten<br>MAK-KZGW: 2.7 mg/m³ 15 Minuten<br>MAK-TMW: 0.5 ppm 8 Stunden<br>MAK-TMW: 1.3 mg/m³ 8 Stunden | TWA: 0.5 ppm 8 timer<br>TWA: 1.3 mg/m³ 8 timer<br>STEL: 2.7 mg/m³ 15 minutter<br>STEL: 1 ppm 15 minutter | STEL: 1 ppm 15 Minuten<br>STEL: 2.7 mg/m³ 15 Minuten<br>TWA: 0.5 ppm 8 Stunden<br>TWA: 1.3 mg/m³ 8 Stunden | STEL: 2.7 mg/m³ 15 minutach<br>TWA: 1.3 mg/m³ 8 godzinach | TWA: 0.5 ppm 8 timer<br>TWA: 1.3 mg/m³ 8 timer<br>STEL: 1 ppm 15 minutter. value from the regulation<br>STEL: 2.7 mg/m³ 15 minutter. value from the regulation |

| Component | Bulgaria | Croatia | Ireland | Cyprus | Czech Republic |
|---|---|---|---|---|---|
| Methanol | TWA: 200 ppm<br>TWA: 260.0 mg/m³<br>Skin notation | kože<br>TWA-GVI: 200 ppm 8 satima.<br>TWA-GVI: 260 mg/m³ 8 satima. | TWA: 200 ppm 8 hr.<br>TWA: 260 mg/m³ 8 hr.<br>STEL: 600 ppm 15 min<br>STEL: 780 mg/m³ 15 min<br>Skin | Skin-potential for cutaneous absorption<br>TWA: 200 ppm<br>TWA: 260 mg/m³ | TWA: 250 mg/m³ 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 1000 mg/m³ |

| | | | | | |
|---|---|---|---|---|---|
| Diethanolamine | TWA: 10 mg/m$^3$ | kože<br>TWA-GVI: 3 ppm 8 satima.<br>TWA-GVI: 15 mg/m$^3$ 8 satima. | TWA: 0.2 ppm 8 hr.<br>TWA: 1 mg/m$^3$ 8 hr. inhalable fraction and vapour<br>STEL: 0.6 ppm 15 min<br>STEL: 3 mg/m$^3$ 15 min<br>Skin | | TWA: 5 mg/m$^3$ 8 hodinách.<br>Ceiling: 10 mg/m$^3$ |
| Ethylene glycol | TWA: 52 mg/m$^3$<br>TWA: 20 ppm<br>STEL : 40 ppm<br>STEL : 104 mg/m$^3$<br>Skin notation | kože<br>TWA-GVI: 20 ppm 8 satima.<br>TWA-GVI: 52 mg/m$^3$ 8 satima.<br>STEL-KGVI: 40 ppm 15 minutama.<br>STEL-KGVI: 104 mg/m$^3$ 15 minutama. | TWA: 20 ppm 8 hr.<br>TWA: 52 mg/m$^3$ 8 hr.<br>STEL: 40 ppm 15 min<br>STEL: 104 mg/m$^3$ 15 min<br>Skin | Skin-potential for cutaneous absorption<br>STEL: 40 ppm<br>STEL: 104 mg/m$^3$<br>TWA: 20 ppm<br>TWA: 52 mg/m$^3$ | TWA: 50 mg/m$^3$ 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 100 mg/m$^3$ |
| Sulfur dioxide | TWA: 1.3 mg/m$^3$<br>TWA: 0.5 ppm<br>STEL : 2.7 mg/m$^3$<br>STEL : 1 ppm | TWA-GVI: 0.5 ppm 8 satima.<br>TWA-GVI: 1.3 mg/m$^3$ 8 satima.<br>STEL-KGVI: 1 ppm 15 minutama.<br>STEL-KGVI: 2.7 mg/m$^3$ 15 minutama. | TWA: 0.5 ppm 8 hr.<br>TWA: 1.3 mg/m$^3$ 8 hr.<br>STEL: 2.7 mg/m$^3$ 15 min<br>STEL: 1 ppm 15 min | STEL: 2.7 mg/m$^3$<br>STEL: 1 ppm<br>TWA: 1.3 mg/m$^3$<br>TWA: 0.5 ppm | TWA: 1.3 mg/m$^3$ 8 hodinách.<br>Ceiling: 2.7 mg/m$^3$ |

| Component | Estonia | Gibraltar | Greece | Hungary | Iceland |
|---|---|---|---|---|---|
| Methanol | Nahk<br>TWA: 200 ppm 8 tundides.<br>TWA: 250 mg/m$^3$ 8 tundides.<br>STEL: 250 ppm 15 minutites.<br>STEL: 350 mg/m$^3$ 15 minutites. | Skin notation<br>TWA: 200 ppm 8 hr<br>TWA: 260 mg/m$^3$ 8 hr | skin - potential for cutaneous absorption<br>STEL: 250 ppm<br>STEL: 325 mg/m$^3$<br>TWA: 200 ppm<br>TWA: 260 mg/m$^3$ | TWA: 260 mg/m$^3$ 8 órában. AK lehetséges borön keresztüli felszívódás | TWA: 200 ppm 8 klukkustundum.<br>TWA: 260 mg/m$^3$ 8 klukkustundum.<br>Skin notation<br>Ceiling: 400 ppm<br>Ceiling: 520 mg/m$^3$ |
| Diethanolamine | Nahk<br>TWA: 3 ppm 8 tundides.<br>TWA: 5 mg/m$^3$ 8 tundides.<br>STEL: 6 ppm 15 minutites.<br>STEL: 30 mg/m$^3$ 15 minutites. | | TWA: 3 ppm<br>TWA: 15 mg/m$^3$ | | TWA: 0.46 ppm 8 klukkustundum.<br>TWA: 2 mg/m$^3$ 8 klukkustundum.<br>Skin notation<br>Ceiling: 0.92 ppm<br>Ceiling: 4 mg/m$^3$ |
| Ethylene glycol | Nahk<br>TWA: 20 ppm 8 tundides. total concentration of aerosol and vapor<br>TWA: 52 mg/m$^3$ 8 tundides. total concentration of aerosol and vapor<br>STEL: 40 ppm 15 minutites. total concentration of aerosol and vapor<br>STEL: 104 mg/m$^3$ 15 minutites. total concentration of aerosol and vapor | Skin notation<br>TWA: 20 ppm 8 hr<br>TWA: 52 mg/m$^3$ 8 hr<br>STEL: 40 ppm 15 min<br>STEL: 104 mg/m$^3$ 15 min | STEL: 50 ppm<br>STEL: 125 mg/m$^3$<br>TWA: 50 ppm<br>TWA: 125 mg/m$^3$ | STEL: 104 mg/m$^3$ 15 percekben. CK<br>TWA: 52 mg/m$^3$ 8 órában. AK lehetséges borön keresztüli felszívódás | STEL: 40 ppm<br>STEL: 104 mg/m$^3$<br>TWA: 10 ppm 8 klukkustundum.<br>TWA: 26 mg/m$^3$ 8 klukkustundum.<br>TWA: 10 ppm 8 klukkustundum. aerosol<br>TWA: 26 mg/m$^3$ 8 klukkustundum. aerosol<br>Skin notation<br>Ceiling: 20 ppm aerosol<br>Ceiling: 52 mg/m$^3$ aerosol |
| Sulfur dioxide | TWA: 0.5 ppm 8 tundides.<br>TWA: 1.3 mg/m$^3$ 8 tundides.<br>STEL: 1 ppm 15 minutites.<br>STEL: 2.7 mg/m$^3$ 15 minutites. | TWA: 1.3 mg/m$^3$ 8 hr<br>TWA: 0.5 ppm 8 hr<br>STEL: 2.7 mg/m$^3$ 15 min<br>STEL: 1 ppm 15 min | STEL: 1 ppm<br>STEL: 2.7 mg/m$^3$<br>TWA: 0.5 ppm<br>TWA: 1.3 mg/m$^3$ | STEL: 2.7 mg/m$^3$ 15 percekben. CK<br>TWA: 1.3 mg/m$^3$ 8 órában. AK | STEL: 1 ppm<br>STEL: 2.7 mg/m$^3$<br>TWA: 0.5 ppm 8 klukkustundum.<br>TWA: 1.3 mg/m$^3$ 8 klukkustundum. |

| Component | Latvia | Lithuania | Luxembourg | Malta | Romania |
|---|---|---|---|---|---|
| Propylene carbonate | TWA: 2 mg/m³ | TWA: 7 mg/m³ IPRD | | | |
| Methanol | skin - potential for cutaneous exposure<br>TWA: 200 ppm<br>TWA: 260 mg/m³ | TWA: 200 ppm IPRD<br>TWA: 260 mg/m³ IPRD<br>Oda | Possibility of significant uptake through the skin<br>TWA: 200 ppm 8 Stunden<br>TWA: 260 mg/m³ 8 Stunden | possibility of significant uptake through the skin<br>TWA: 200 ppm<br>TWA: 260 mg/m³ | Skin notation<br>TWA: 200 ppm 8 ore<br>TWA: 260 mg/m³ 8 ore |
| Diethanolamine | | TWA: 3 ppm IPRD<br>TWA: 15 mg/m³ IPRD<br>Oda<br>STEL: 6 ppm<br>STEL: 30 mg/m³ | | | |
| Ethylene glycol | skin - potential for cutaneous exposure<br>STEL: 40 ppm<br>STEL: 104 mg/m³<br>TWA: 20 ppm<br>TWA: 52 mg/m³ | TWA: 10 ppm aerosol and vapor IPRD<br>TWA: 25 mg/m³ aerosol and vapor IPRD<br>Oda<br>STEL: 20 ppm<br>STEL: 50 mg/m³ | Possibility of significant uptake through the skin<br>TWA: 20 ppm 8 Stunden<br>TWA: 52 mg/m³ 8 Stunden<br>STEL: 40 ppm 15 Minuten<br>STEL: 104 mg/m³ 15 Minuten | possibility of significant uptake through the skin<br>TWA: 20 ppm<br>TWA: 52 mg/m³<br>STEL: 40 ppm 15 minuti<br>STEL: 104 mg/m³ 15 minuti | Skin notation<br>TWA: 20 ppm 8 ore<br>TWA: 52 mg/m³ 8 ore<br>STEL: 40 ppm 15 minute<br>STEL: 104 mg/m³ 15 minute |
| Sulfur dioxide | STEL: 2.7 mg/m³<br>STEL: 1 ppm<br>TWA: 1.3 mg/m³<br>TWA: 0.5 ppm | TWA: 1.3 mg/m³ IPRD<br>TWA: 0.5 ppm IPRD<br>STEL: 2.7 mg/m³<br>STEL: 1 ppm | TWA: 1.3 mg/m³ 8 Stunden<br>TWA: 0.5 ppm 8 Stunden<br>STEL: 2.7 mg/m³ 15 Minuten<br>STEL: 1 ppm 15 Minuten | TWA: 0.5 ppm<br>TWA: 1.3 mg/m³<br>STEL: 1 ppm 15 minuti<br>STEL: 2.7 mg/m³ 15 minuti | TWA: 0.5 ppm 8 ore<br>TWA: 1.3 mg/m³ 8 ore<br>STEL: 1 ppm 15 minute<br>STEL: 2.7 mg/m³ 15 minute |

| Component | Russia | Slovak Republic | Slovenia | Sweden | Turkey |
|---|---|---|---|---|---|
| Propylene carbonate | MAC: 7 mg/m³ | | | | |
| Methanol | TWA: 5 mg/m³ 1250<br>Skin notation<br>MAC: 15 mg/m³ | Potential for cutaneous absorption<br>TWA: 200 ppm<br>TWA: 260 mg/m³ | TWA: 200 ppm 8 urah<br>TWA: 260 mg/m³ 8 urah<br>Koža<br>STEL: 800 ppm 15 minutah<br>STEL: 1040 mg/m³ 15 minutah | Indicative STEL: 250 ppm 15 minuter<br>Indicative STEL: 350 mg/m³ 15 minuter<br>TLV: 200 ppm 8 timmar. NGV<br>TLV: 250 mg/m³ 8 timmar. NGV<br>Hud | Deri<br>TWA: 200 ppm 8 saat<br>TWA: 260 mg/m³ 8 saat |
| Diethanolamine | Skin notation<br>MAC: 5 mg/m³ | | TWA: 0.5 mg/m³ 8 urah<br>TWA: 0.11 ppm 8 urah<br>Koža<br>STEL: 0.11 ppm 15 minutah<br>STEL: 0.5 mg/m³ 15 minutah | Indicative STEL: 6 ppm 15 minuter<br>Indicative STEL: 30 mg/m³ 15 minuter<br>TLV: 3 ppm 8 timmar. NGV<br>TLV: 15 mg/m³ 8 timmar. NGV<br>Hud | |
| Ethylene glycol | TWA: 5 mg/m³ 2388<br>MAC: 10 mg/m³ | Ceiling: 104 mg/m³<br>Potential for cutaneous absorption<br>TWA: 20 ppm<br>TWA: 52 mg/m³ | TWA: 20 ppm 8 urah<br>TWA: 52 mg/m³ 8 urah<br>Koža<br>STEL: 40 ppm 15 minutah<br>STEL: 104 mg/m³ 15 minutah | Binding STEL: 40 ppm 15 minuter<br>Binding STEL: 104 mg/m³ 15 minuter<br>TLV: 10 ppm 8 timmar. NGV<br>TLV: 25 mg/m³ 8 timmar. NGV<br>Hud | Deri<br>TWA: 20 ppm 8 saat<br>TWA: 52 mg/m³ 8 saat<br>STEL: 40 ppm 15 dakika<br>STEL: 104 mg/m³ 15 dakika |
| Sulfur dioxide | Skin notation<br>MAC: 10 mg/m³ | Ceiling: 2.7 mg/m³<br>TWA: 0.5 ppm<br>TWA: 1.3 mg/m³ | TWA: 0.5 ppm 8 urah<br>TWA: 1.3 mg/m³ 8 urah<br>STEL: 1 ppm 15 minutah<br>STEL: 2.7 mg/m³ 15 minutah | Binding STEL: 1 ppm 15 minuter<br>Binding STEL: 2.7 mg/m³ 15 minuter<br>TLV: 0.5 ppm 8 timmar. NGV<br>TLV: 1.3 mg/m³ 8 timmar. NGV | |

**Biological limit values**
List source(s):

| Component | European Union | United Kingdom | France | Spain | Germany |
|---|---|---|---|---|---|
| Methanol | | | Methanol: 15 mg/L urine end of shift | Methanol: 15 mg/L urine end of shift | Methanol: 15 mg/L urine (end of shift ) Methanol: 15 mg/L urine (for long-term exposures: at the end of the shift after several shifts ) |

| Component | Italy | Finland | Denmark | Bulgaria | Romania |
|---|---|---|---|---|---|
| Methanol | | | | | Methanol: 6 mg/L urine end of shift |

| Component | Gibraltar | Latvia | Slovak Republic | Luxembourg | Turkey |
|---|---|---|---|---|---|
| Methanol | | | Methanol: 30 mg/L urine end of exposure or work shift Methanol: 30 mg/L urine after all work shifts for long-term exposure | | |

**Monitoring methods**
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS70 General methods for sampling airborne gases and vapours
MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography
MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

**Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)**
See table for values

| Component | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|---|---|---|---|---|
| Propylene carbonate 108-32-7 ( 38 ) | | | DNEL = 10mg/cm2 | DNEL = 20mg/kg bw/day |
| Methanol 67-56-1 ( 30 ) | | DNEL = 20mg/kg bw/day | | DNEL = 20mg/kg bw/day |
| Diethanolamine 111-42-2 ( 12 ) | | | | DNEL = 0.13mg/kg bw/day |
| Ethylene glycol 107-21-1 ( 8 ) | | | | DNEL = 106mg/kg bw/day |
| 1-Imidazole 288-32-4 ( 3 ) | | | | DNEL = 1.5mg/kg bw/day |

| Component | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|---|---|---|---|---|
| Propylene carbonate 108-32-7 ( 38 ) | | | DNEL = 20mg/m³ | DNEL = 70.53mg/m³ |
| Methanol 67-56-1 ( 30 ) | DNEL = 130mg/m³ | DNEL = 130mg/m³ | DNEL = 130mg/m³ | DNEL = 130mg/m³ |
| Diethanolamine 111-42-2 ( 12 ) | | | DNEL = 0.5mg/m³ | DNEL = 0.75mg/m³ |
| Ethylene glycol | | | DNEL = 35mg/m³ | DNEL = 70mg/m³ |

| | | | | |
|---|---|---|---|---|
| 107-21-1 ( 8 ) | | | | |
| Sulfur dioxide<br>7446-09-5 ( 7 ) | DNEL = 2.7mg/m³ | | DNEL = 2.7mg/m³ | |
| 1-Imidazole<br>288-32-4 ( 3 ) | | | | DNEL = 10.6mg/m³ |

**Predicted No Effect Concentration (PNEC)**
See values below.

| Component | Fresh water | Fresh water sediment | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture) |
|---|---|---|---|---|---|
| Propylene carbonate<br>108-32-7 ( 38 ) | PNEC = 0.9mg/L | | PNEC = 9mg/L | PNEC = 7400mg/L | PNEC = 0.81mg/kg soil dw |
| Methanol<br>67-56-1 ( 30 ) | PNEC = 20.8mg/L | PNEC = 77mg/kg sediment dw | PNEC = 1540mg/L | PNEC = 100mg/L | PNEC = 100mg/kg soil dw |
| Diethanolamine<br>111-42-2 ( 12 ) | PNEC = 0.021mg/L | PNEC = 0.092mg/kg sediment dw | PNEC = 0.095mg/L | PNEC = 100mg/L | PNEC = 1.63mg/kg soil dw |
| Ethylene glycol<br>107-21-1 ( 8 ) | PNEC = 10mg/L | PNEC = 20.9 mg/kg sediment dw | PNEC = 10mg/L | PNEC = 199.5mg/L | PNEC = 1.53mg/kg soil dw |
| 1-Imidazole<br>288-32-4 ( 3 ) | PNEC = 0.13mg/L | PNEC = 0.336mg/kg sediment dw | PNEC = 1.3mg/L | PNEC = 10mg/L | PNEC = 0.0425mg/kg soil dw |

| Component | Marine water | Marine water sediment | Marine water Intermittent | Food chain | Air |
|---|---|---|---|---|---|
| Propylene carbonate<br>108-32-7 ( 38 ) | PNEC = 0.09mg/L | | PNEC = 0.9mg/L | | |
| Methanol<br>67-56-1 ( 30 ) | PNEC = 2.08mg/L | PNEC = 7.7mg/kg sediment dw | | | |
| Diethanolamine<br>111-42-2 ( 12 ) | PNEC = 0.002mg/L | PNEC = 0.0092mg/kg sediment dw | | PNEC = 1.04mg/kg food | |
| Ethylene glycol<br>107-21-1 ( 8 ) | PNEC = 1mg/L | PNEC = 3.7mg/kg sediment dw<br>PNEC = 31.2mg/kg sediment dw<br>PNEC = 31.7mg/kg sediment dw | PNEC = 10mg/L | | |
| 1-Imidazole<br>288-32-4 ( 3 ) | PNEC = 0.013mg/L | PNEC = 0.0336mg/kg sediment dw | | | |

## 8.2. Exposure controls

**Engineering Measures**
Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments |
|---|---|---|---|---|
| Nitrile rubber | See manufacturers | - | | (minimum requirement) |

| | | |
|---|---|---|
| Neoprene<br>Natural rubber<br>PVC | recommendations | EN 374 |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| **Physical State** | Liquid | |
| **Appearance** | | |
| **Odor** | No information available | |
| **Odor Threshold** | No data available | |
| **Melting Point/Range** | No data available | |
| **Softening Point** | No data available | |
| **Boiling Point/Range** | No information available | |
| **Flammability (liquid)** | Flammable | On basis of test data Estimated |
| **Flammability (solid,gas)** | Not applicable | Liquid |
| **Explosion Limits** | No data available | |
| **Flash Point** | No information available °C / °F | **Method -** No information available |
| **Autoignition Temperature** | No data available | |
| **Decomposition Temperature** | No data available | |
| **pH** | Not applicable | |
| **Viscosity** | No data available | |
| **Water Solubility** | No information available | |
| **Solubility in other solvents** | No information available | |

**Partition Coefficient (n-octanol/water)**

| Component | log Pow |
|---|---|
| Propylene carbonate | -0.5 |
| Methanol | -0.74 |
| Diethanolamine | -2.46 |
| Ethylene glycol | -1.36 |
| 1-Imidazole | -0.02 |

| | | |
|---|---|---|
| **Vapor Pressure** | No data available | |
| **Density / Specific Gravity** | 1.2 | |
| **Bulk Density** | Not applicable | Liquid |
| **Vapor Density** | No data available | (Air = 1.0) |
| **Particle characteristics** | Not applicable (liquid) | |

### 9.2. Other information

explosive air/vapour mixtures possible

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

**Hazardous Polymerization** No information available.
**Hazardous Reactions** None under normal processing.

### 10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

None known.

### 10.6. Hazardous decomposition products

Carbon oxides. Nitrogen oxides (NOx). Sulfur oxides. Hydrogen iodide.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

**Product Information**

**(a) acute toxicity;**
- **Oral** Category 3
- **Dermal** Category 4
- **Inhalation** Category 3

### Toxicology data for the components

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|---|---|---|---|
| Propylene carbonate | LD50 = 29000 mg/kg ( Rat ) | LD50 > 3000 mg/kg ( Rabbit ) | - |
| Methanol | LD50 = 1187 – 2769 mg/kg (Rat) | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h |
| Diethanolamine | LD50 = 780 mg/kg ( Rat ) | LD50 = 11.9 mL/kg ( Rabbit ) | - |
| Ethylene glycol | LD50 = 4700 mg/kg ( Rat ) | LD50 = 10600 mg/kg ( Rat ) | LC50 > 2.5 mg/L ( Rat ) 6 h |
| Sulfur dioxide | - | - | Per CGA P-20: 2500 ppm/1hr ( Rat ) |
| 1-Imidazole | 970 mg/kg (Rat) | - | - |

**(b) skin corrosion/irritation;** Category 1 B

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**
**Respiratory** No data available
**Skin** No data available

| Component | Test method | Test species | Study result |
| --- | --- | --- | --- |
| Methanol<br>67-56-1 ( 30 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test (GPMT) | guinea pig | non-sensitising |

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC |
| --- | --- | --- | --- | --- |
| Diethanolamine | | | | Group 2B |

**(g) reproductive toxicity;** Category 1B

| Component | Test method | Test species / Duration | Study result |
| --- | --- | --- | --- |
| Methanol<br>67-56-1 ( 30 ) | OECD Test Guideline 416 | Rat / Inhalation<br>2 Generation | NOAEC =<br>1.3 mg/l (air) |

**(h) STOT-single exposure;** Category 1

**Results / Target organs** Optic nerve, Central nervous system (CNS).

**(i) STOT-repeated exposure;** Category 2

**Target Organs** No information available.

**(j) aspiration hazard;** No data available

**Symptoms / effects,both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

**11.2. Information on other hazards**

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**12.1. Toxicity**
**Ecotoxicity effects** Contains a substance which is:. Toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Fish | Water Flea | Freshwater Algae |
| --- | --- | --- | --- |

| | | | |
|---|---|---|---|
| Propylene carbonate | Leuciscus idus: LC50: 5300 mg/L/96h | EC50: > 500 mg/L, 48h (Daphnia magna) | EC50: > 500 mg/L, 72h (Desmodesmus subspicatus) |
| Methanol | Pimephales promelas: LC50 > 10000 mg/L 96h | EC50 > 10000 mg/L 24h | |
| Diethanolamine | Pimephals prome: LC50: 140 mg/L/96h | EC50: = 55 mg/L, 48h (Daphnia magna) | EC50: 2.1 - 2.3 mg/L, 96h (Pseudokirchneriella subcapitata)<br>EC50: = 7.8 mg/L, 72h (Desmodesmus subspicatus) |
| Ethylene glycol | LC50: = 41000 mg/L, 96h (Oncorhynchus mykiss)<br>LC50: = 27540 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 14 - 18 mL/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 40761 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 40000 - 60000 mg/L, 96h static (Pimephales promelas)<br>LC50: = 16000 mg/L, 96h static (Poecilia reticulata) | EC50: = 46300 mg/L, 48h (Daphnia magna) | EC50: 6500 - 13000 mg/L, 96h (Pseudokirchneriella subcapitata) |
| 1-Imidazole | | EC50: = 341.5 mg/L, 48h (Daphnia magna) | EC50: = 82 mg/L, 96h (Desmodesmus subspicatus)<br>EC50: = 130 mg/L, 72h (Desmodesmus subspicatus) |

| Component | Microtox | M-Factor |
|---|---|---|
| Propylene carbonate | EC50 > 10000 mg/L 17 h | |
| Methanol | EC50 = 39000 mg/L 25 min<br>EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min | |
| Diethanolamine | EC50 = 73 mg/L 5 min<br>EC50 > 16 mg/L 16 h | |
| Ethylene glycol | EC50 = 10000 mg/L 16 h<br>EC50 = 620 mg/L 30 min<br>EC50 = 620.0 mg/L 30 min | |
| 1-Imidazole | = 1200 mg/L EC50 Pseudomonas putida 17 h<br>= 231 mg/L EC50 Photobacterium phosphoreum 30 min | |

**12.2. Persistence and degradability** No information available

| Component | Degradability |
|---|---|
| Methanol<br>67-56-1 ( 30 ) | DT50 ~ 17.2d<br>>94% after 20d |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**12.3. Bioaccumulative potential** No information available

| Component | log Pow | Bioconcentration factor (BCF) |
|---|---|---|
| Propylene carbonate | -0.5 | No data available |
| Methanol | -0.74 | <10 dimensionless |
| Diethanolamine | -2.46 | No data available |
| Ethylene glycol | -1.36 | No data available |
| 1-Imidazole | -0.02 | No data available |

**12.4. Mobility in soil** No information available

**12.5. Results of PBT and vPvB** No data available for assessment.

**assessment**

**12.6. Endocrine disrupting properties**

| | |
|---|---|
| **Endocrine Disruptor Information** | This product does not contain any known or suspected endocrine disruptors |

**12.7. Other adverse effects**

| | |
|---|---|
| **Persistent Organic Pollutant** | This product does not contain any known or suspected substance |
| **Ozone Depletion Potential** | This product does not contain any known or suspected substance |

## SECTION 13: DISPOSAL CONSIDERATIONS

**13.1. Waste treatment methods**

| | |
|---|---|
| **Waste from Residues/Unused Products** | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| **Contaminated Packaging** | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition. |
| **European Waste Catalogue (EWC)** | According to the European Waste Catalog, Waste Codes are not product specific, but application specific. |
| **Other Information** | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |
| **Switzerland - Waste Ordinance** | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br>https://www.fedlex.admin.ch/eli/cc/2015/891/en |

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**

| | |
|---|---|
| **14.1. UN number** | UN2924 |
| **14.2. UN proper shipping name** | FLAMMABLE LIQUID, CORROSIVE, N.O.S. |
| **Technical Shipping Name** | Methanol/Sulfur dioxide |
| **14.3. Transport hazard class(es)** | 3 |
| **Subsidiary Hazard Class** | 8 |
| **14.4. Packing group** | III |

**ADR**

| | |
|---|---|
| **14.1. UN number** | UN2924 |
| **14.2. UN proper shipping name** | FLAMMABLE LIQUID, CORROSIVE, N.O.S. |
| **Technical Shipping Name** | Methanol/Sulfur dioxide |
| **14.3. Transport hazard class(es)** | 3 |
| **Subsidiary Hazard Class** | 8 |
| **14.4. Packing group** | III |

**IATA**

| | |
|---|---|
| **14.1. UN number** | UN2924 |

**14.2. UN proper shipping name** FLAMMABLE LIQUID, CORROSIVE, N.O.S.
**Technical Shipping Name** Methanol/Sulfur dioxide
**14.3. Transport hazard class(es)** 3
**Subsidiary Hazard Class** 8
**14.4. Packing group** III

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No | EINECS | ELINCS | NLP | IECSC | TCSI | KECL | ENCS | ISHL |
|---|---|---|---|---|---|---|---|---|---|
| Propylene carbonate | 108-32-7 | 203-572-1 | - | - | X | X | KE-23785 | X | X |
| Methanol | 67-56-1 | 200-659-6 | - | - | X | X | KE-23193 | X | X |
| Diethanolamine | 111-42-2 | 203-868-0 | - | - | X | X | KE-20959 | X | X |
| Ethylene glycol | 107-21-1 | 203-473-3 | - | - | X | X | KE-13169 | X | X |
| Sulfur dioxide | 7446-09-5 | 231-195-2 | - | - | X | X | KE-32567 | X | X |
| 1-Imidazole | 288-32-4 | 206-019-2 | - | - | X | X | KE-20937 | X | X |
| Iodized salt | N/A | - | - | - | - | - | - | - | - |

| Component | CAS No | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|---|---|---|---|---|---|---|---|---|
| Propylene carbonate | 108-32-7 | X | ACTIVE | X | - | X | X | X |
| Methanol | 67-56-1 | X | ACTIVE | X | - | X | X | X |
| Diethanolamine | 111-42-2 | X | ACTIVE | X | - | X | X | X |
| Ethylene glycol | 107-21-1 | X | ACTIVE | X | - | X | X | X |
| Sulfur dioxide | 7446-09-5 | X | ACTIVE | X | - | X | X | X |
| 1-Imidazole | 288-32-4 | X | ACTIVE | X | - | X | X | X |
| Iodized salt | N/A | - | - | - | - | - | - | - |

**Legend:** X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

#### Authorisation/Restrictions according to EU REACH

| Component | CAS No | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---|---|---|---|---|
| Propylene carbonate | 108-32-7 | - | Use restricted. See item 75.<br>(see link for restriction details) | - |
| Methanol | 67-56-1 | - | Use restricted. See item 69.<br>(see link for restriction details)<br>Use restricted. See item 75. | - |

| | | | | |
|---|---|---|---|---|
| | | | (see link for restriction details) | |
| Diethanolamine | 111-42-2 | - | Use restricted. See item 75.<br>(see link for restriction details) | - |
| Ethylene glycol | 107-21-1 | - | - | - |
| Sulfur dioxide | 7446-09-5 | - | Use restricted. See item 75.<br>(see link for restriction details) | - |
| 1-Imidazole | 288-32-4 | - | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | - |
| Iodized salt | N/A | - | - | - |

**REACH links**
*https://echa.europa.eu/substances-restricted-under-reach*

**Seveso III Directive (2012/18/EC)**

| Component | CAS No | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---|---|---|---|
| Propylene carbonate | 108-32-7 | Not applicable | Not applicable |
| Methanol | 67-56-1 | 500 tonne | 5000 tonne |
| Diethanolamine | 111-42-2 | Not applicable | Not applicable |
| Ethylene glycol | 107-21-1 | Not applicable | Not applicable |
| Sulfur dioxide | 7446-09-5 | Not applicable | Not applicable |
| 1-Imidazole | 288-32-4 | Not applicable | Not applicable |
| Iodized salt | N/A | Not applicable | Not applicable |

**Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals**
Not applicable

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
Take note of Directive 94/33/EC on the protection of young people at work
Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

**National Regulations**

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** Water endangering class = 2 (self classification)

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|---|---|---|
| Propylene carbonate | WGK1 | |
| Methanol | WGK 2 | Class I : 20 mg/m³ (Massenkonzentration) |
| Diethanolamine | WGK2 | |
| Ethylene glycol | WGK1 | |

| | | |
|---|---|---|
| Sulfur dioxide | WGK1 | |
| 1-Imidazole | WGK2 | |

| Component | France - INRS (Tables of occupational diseases) |
|---|---|
| Methanol | Tableaux des maladies professionnelles (TMP) - RG 84 |
| Diethanolamine | Tableaux des maladies professionnelles (TMP) - RG 49,RG 49bis |
| Ethylene glycol | Tableaux des maladies professionnelles (TMP) - RG 84 |

**Swiss Regulations**
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|---|---|---|---|
| Methanol<br>67-56-1 ( 30 ) | Prohibited and Restricted Substances | Group I | |
| Ethylene glycol<br>107-21-1 ( 8 ) | Prohibited and Restricted Substances | | |

### 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

**Full text of H-Statements referred to under sections 2 and 3**
H301 - Toxic if swallowed
H312 - Harmful in contact with skin
H331 - Toxic if inhaled
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H370 - Causes damage to organs
H360D - May damage the unborn child
H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child
H373 - May cause damage to organs through prolonged or repeated exposure
H225 - Highly flammable liquid and vapor
H302 - Harmful if swallowed
H311 - Toxic in contact with skin
H315 - Causes skin irritation
H319 - Causes serious eye irritation

**Legend**

**CAS** - Chemical Abstracts Service
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances
**IECSC** - Chinese Inventory of Existing Chemical Substances
**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List
**ENCS** - Japanese Existing and New Chemical Substances
**AICS** - Australian Inventory of Chemical Substances
**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit
**ACGIH** - American Conference of Governmental Industrial Hygienists
**DNEL** - Derived No Effect Level
**RPE** - Respiratory Protective Equipment
**LC50** - Lethal Concentration 50%
**NOEC** - No Observed Effect Concentration

**TWA** - Time Weighted Average
**IARC** - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
**LD50** - Lethal Dose 50%
**EC50** - Effective Concentration 50%
**POW** - Partition coefficient Octanol:Water

**PBT** - Persistent, Bioaccumulative, Toxic
**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code
**OECD** - Organisation for Economic Co-operation and Development
**BCF** - Bioconcentration factor
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association
**MARPOL** - International Convention for the Prevention of Pollution from Ships
**ATE** - Acute Toxicity Estimate
**VOC** - (volatile organic compound)

**Key literature references and sources for data**
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

| | |
|---|---|
| **Physical hazards** | On basis of test data |
| **Health Hazards** | Calculation method |
| **Environmental hazards** | Calculation method |

**Training Advice**
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.

| | |
|---|---|
| **Prepared By** | Health, Safety and Environmental Department |
| **Creation Date** | 12-Apr-2023 |
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| **Revision Summary** | New emergency telephone response service provider. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006** .

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**



## End of Safety Data Sheet